

Solutions to Final Examination

- 1.(a) True. If $x = \sqrt{\sqrt{2}-1}$ were rational, then so would be $\sqrt{2} = x^2 + 1$. We know that this is false. 2
- (b) True. If A is a countable subset of $\mathbb{R}$, then the elements of A can be listed as a sequence that takes on each value exactly once. According to the Bolzano-Weierstrass Theorem, this sequence has a convergent subsequence. The limit of this subsequence will be a limit point of A . 3
- (c) False. Let f be the constant function defined by $f(x) = \sqrt{2}$ for all $x \in [a, b]$. Then f is continuous on $[a, b]$, but $f(x)$ is never rational. 2

- 2.(a) We use induction. First note that $a_1 = 3$ and $a_2 = 5$, so the inequalities hold in the case when $k = 1$. Suppose now that

$$2 \leq a_n \leq a_{n+1} \leq 27$$

for some $n \in \mathbb{N}$. Then

$$2 + \sqrt{6} \leq 2 + \sqrt{3a_n} \leq 2 + \sqrt{3a_{n+1}} \leq 11,$$

so

$$2 \leq 2 + \sqrt{6} \leq a_{n+1} \leq a_{n+2} \leq 11 \leq 27.$$

The Principle of Mathematical Induction tells us that the inequalities hold for all $k \in \mathbb{N}$. 5

- (b) The *Monotone Convergence Theorem* says that if a sequence is increasing and bounded above, then it converges (to its supremum). Since we have shown $a_k \leq a_{k+1}$ for all $k \in \mathbb{N}$, the sequence $(a_k)_{k=1}^{\infty}$ is increasing. We have also proved that $(a_k)_{k=1}^{\infty}$ is bounded above by 27. 4
- (c) Put $x = \lim_{k \rightarrow \infty} a_k$. From $a_{k+1} = 2 + \sqrt{3a_k}$, we deduce that $(a_{k+1} - 2)^2 = 3a_k$. Taking limits on both sides of this equation, we get

$$(x - 2)^2 = 3x \iff x^2 - 7x + 4 = 0.$$

The quadratic formula says that this has two solutions, $x = \frac{7 \pm \sqrt{33}}{2}$. Since $\frac{7}{2} - \frac{1}{2}\sqrt{33} < 2 \leq a_k$ for all k , we conclude that $\lim_{k \rightarrow \infty} a_k = \frac{7}{2} + \frac{1}{2}\sqrt{33}$. 6

3. First note that

$$\left| \frac{2k + b_k}{k + 1} - 2 \right| = \frac{|b_k - 2|}{k + 1}.$$

Since $\lim_{k \rightarrow \infty} b_k = 3$, we can find an $N_1 \in \mathbb{N}$ such that $|b_k - 3| < 1$ for all integers $k \geq N_1$. For such k ,

$$|b_k - 2| = |b_k - 3 + 1| \leq |b_k - 3| + 1 < 2.$$

Given any $\varepsilon > 0$, choose an $N \in \mathbb{N}$ so that $N \geq N_1$ and $N > 2/\varepsilon$. For any integer k with $k \geq N$, we have that $|b_k - 2| < 2$ and $2/k \leq 2/N < \varepsilon$, so

$$\left| \frac{2k + b_k}{k + 1} - 2 \right| < \frac{2}{k + 1} < \frac{2}{k} < \varepsilon.$$

This proves that $\lim_{k \rightarrow \infty} \frac{2k + b_k}{k + 1} = 2$. 9

4.(a) We use the Ratio Test: put $L = \lim_{k \rightarrow \infty} \frac{(k+1)\left(\frac{2}{3}\right)^{k+1}}{k\left(\frac{2}{3}\right)^k} = \frac{2}{3}$. Then $L < 1$, so the Ratio Test says that the series converges.

(b) Note that

$$0 < \frac{k + \ln(k)}{k^3 + k + 1} < \frac{k + k}{k^3 + 0 + 0} = 2 \frac{1}{k^2}$$

for all $k \in \mathbb{N}$. Since $\sum_{k=1}^{\infty} \frac{1}{k^2}$ is a convergent p -series, the Comparison Test says that $\sum_{k=1}^{\infty} \frac{k + \ln(k)}{k^3 + k + 1}$ converges.

(c) $\lim_{k \rightarrow \infty} 3^{1/k} = 1$, so the series diverges.

5.(a) Since $0 < \ln(k) < k$ for all $k > 1$,

$$0 < \frac{1}{k} < \frac{1}{\ln(k)}$$

for all $k \geq 2$. Since $\sum_{k=2}^{\infty} \frac{1}{k}$ is a harmonic series, and therefore diverges, the Comparison Test says

that the series $\sum_{k=2}^{\infty} \frac{1}{\ln(k)}$ diverges. Thus the given series $\sum_{k=2}^{\infty} \frac{(-1)^k}{\ln(k)}$ is not absolutely convergent.

However, the function $\ln(x)$ is strictly increasing and $\ln(x) \rightarrow \infty$ as $x \rightarrow \infty$, so the Alternating Series Test tells us that $\sum_{k=2}^{\infty} \frac{(-1)^k}{\ln(k)}$ is conditionally convergent.

(b) Since $L = \lim_{k \rightarrow \infty} \frac{1}{\ln(k)} = 0$, the Root Test tells us that $\sum_{k=2}^{\infty} \left(\frac{-1}{\ln(k)}\right)^k$ is absolutely convergent.

6.(a) If $x > 0$, then $\lim_{k \rightarrow \infty} h_k(x) = \lim_{k \rightarrow \infty} e^{-kx} = 0$. However, if $x = 0$, then $h_k(x) = 1$ for all $k \in \mathbb{N}$. Thus

$$h(x) = \begin{cases} 0 & \text{if } x > 0, \\ 1 & \text{if } x = 0. \end{cases}$$

(b) The uniform limit of a sequence of continuous functions must be continuous. Since each of the h_k 's is continuous but h is not continuous at $x = 0$, we conclude that the convergence cannot be uniform.

Alternatively, you can note that $h_k(1/k) = e^{-1}$ for all $k \in \mathbb{N}$. This means that for any $\delta > 0$ and for every $N \in \mathbb{N}$, there is a $k \geq N$ and an $x \geq 0$ such that $|x| < \delta$ and $|h_k(x)| \geq e^{-1}$.

7.(a) $R = \lim_{k \rightarrow \infty} \frac{(k+1)^4 4^{k+1}}{k^4 4^k} = 4.$

(b) $R = \lim_{k \rightarrow \infty} \frac{2^k}{3^k + (-1)^k 4k} \cdot \frac{3^{k+1} - (-1)^k 4(k+1)}{2^{k+1}} = \frac{3}{2}.$

8.(a) K is compact if every sequence in K has a subsequence that converges to a point in K .

(b) Since $y_n \in f^{-1}(K)$, there must be at least one point $x_n \in K$ with $f(x_n) = y_n$. ✓

(c) The definition of a continuous function says that for every $\varepsilon > 0$, we can find a $\delta > 0$ such that ✓

$$(\forall y \in \mathbb{R}) \quad |y - x| < \delta \implies |f(y) - f(x)| < \varepsilon. \quad \checkmark$$

Because the subsequence $(x_{n_k})_{k=1}^{\infty}$ converges to x , there is an $N \in \mathbb{N}$ such that ✓

$$\begin{aligned} (\forall k \in \mathbb{N}) \quad k \geq N &\implies |x_{n_k} - x| < \delta \quad \checkmark \\ &\implies |f(x_{n_k}) - f(x)| < \varepsilon \quad \checkmark \\ &\implies |y_{n_k} - f(x)| < \varepsilon. \quad \checkmark \end{aligned}$$

Thus $\lim_{k \rightarrow \infty} y_{n_k} = f(x)$.

(d) Since $x \in K$, $f(x) \in f^{-1}(K)$. We have shown that any sequence in $f^{-1}(K)$ has a subsequence that converges to a point in $f^{-1}(K)$. Thus $f^{-1}(K)$ is compact. ✓

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